

FEATURE LEARNING · DEPTH & HIERARCHY · TRAINING DYNAMICS · SCALING LAWS · SPIN GLASSES

## SUMMARY

My research asks which structures make high-dimensional problems statistically or algorithmically hard, and which permit efficient learning or computation. In particular, I develop predictive theories of feature learning and the role of depth in neural networks, together with theories of dynamics, optimization, and statistical-to-computational gaps in spin glasses and random graphs. This work draws on mathematical tools from probability, optimization, learning theory, and statistical physics.

## RESEARCH HIGHLIGHTS

### FEATURE LEARNING

#### Asymptotics of feature learning in two-layer networks

Developed a new route to universality and exact high-dimensional asymptotics by conditioning on gradient-induced spikes, yielding precise characterizations of generalization and spectrum of learned features. Established how batch size and successive gradient steps govern the acquisition of target directions, and showed that reusing batches makes targets learnable that online-SGD analyses classify as hard. These works motivated a refined classification through symmetries and label transformations, and uncovered novel statistical-to-computational gaps.

ICML 2024 Spotlight · AISTATS 2025 Oral · JMLR 2024 · ICML 2024 · AISTATS 2025

### DEPTH & HIERARCHY

#### Depth as sequential dimension reduction

Led work identifying a computational advantage of depth: each layer filters its input into a richer feature space of lower effective dimension, making new features easier to discover. Follow-ups cover hierarchical and compositional targets, scaling laws, and low-degree spectral filtering.

NeurIPS 2025 Spotlight · ICML 2026 · COLT 2026 · HiLD @ ICML 2026

### SPIN GLASSES

#### Statics, dynamics, and statistical-to-computational gaps in spin glasses

Proved exact asymptotic equations governing Glauber dynamics in the SK model, then developed a rigorous dynamical cavity framework for first-order algorithms. Characterized stable local optima in random graphs and spin glasses, probing statistical-to-computational gaps.

COLT 2026 · arXiv 2025 · SLMATH highlight

## SELECTED PUBLICATIONS

Full list of publications: [Google Scholar](#)

### PREPRINT

2026 · Under review

#### Deep Learning as Neural Low-Degree Filtering: A Spectral Theory of Hierarchical Feature Learning [arXiv](#) · [code](#)

Yatin Dandi, Matteo Vilucchio, Luca Arnaboldi, Hugo Tabanelli, Florent Krzakala

### COLT

2026

#### Rigorous Asymptotics for First-Order Algorithms Through the Dynamical Cavity Method [arXiv](#)

Yatin Dandi, David Gamarnik, Francisco Pernice, Lenka Zdeborová

### NEURIPS

2025 · Spotlight

#### The Computational Advantage of Depth: Learning High-Dimensional Hierarchical Functions with Gradient Descent [arXiv](#)

Yatin Dandi, Luca Pesce, Lenka Zdeborová, Florent Krzakala

### PREPRINT

2025 · Under review

#### Sequential Dynamics in Ising Spin Glasses [arXiv](#) · [SLMATH](#)

Yatin Dandi, David Gamarnik, Francisco Pernice, Lenka Zdeborová

### AISTATS

2025 · Oral

#### A Random Matrix Theory Perspective on the Spectrum of Learned Features and Asymptotic Generalization [arXiv](#)

Yatin Dandi, Luca Pesce, Hugo Cui, Florent Krzakala, Yue M. Lu, Bruno Loureiro

### AISTATS

2025

#### Fundamental Computational Limits of Weak Learnability in High-Dimensional Multi-Index Models [paper](#)

Emanuele Troiani, Yatin Dandi, Leonardo Defilippis, Lenka Zdeborová, Bruno Loureiro, Florent Krzakala

### ICML

2024

#### The Benefits of Reusing Batches for Gradient Descent in Two-Layer Networks [arXiv](#)

Yatin Dandi, Emanuele Troiani, Luca Arnaboldi, Luca Pesce, Lenka Zdeborová, Florent Krzakala

|                                 |                                                                                                                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ICML<br>2024 · Spotlight        | <b>Asymptotics of Feature Learning in Two-Layer Networks after One Gradient Step</b> <a href="#">arXiv</a><br>Hugo Cui, Luca Pesce, <b>Yatin Dandi</b> , Florent Krzakala, Yue M. Lu, Lenka Zdeborová, Bruno Loureiro |
| JMLR<br>2024                    | <b>How Two-Layer Neural Networks Learn, One (Giant) Step at a Time</b> <a href="#">paper</a> · <a href="#">arXiv</a><br><b>Yatin Dandi</b> , Florent Krzakala, Bruno Loureiro, Luca Pesce, Ludovic Stephan            |
| PREPRINT<br>2023 · Under review | <b>Maximally-Stable Local Optima in Random Graphs and Spin Glasses: Phase Transitions and Universality</b> <a href="#">arXiv</a><br><b>Yatin Dandi</b> , David Gamarnik, Lenka Zdeborová                              |

## RESEARCH EXPERIENCE

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SINCE 2022  | <b>PhD Researcher</b><br><i>EPFL · Advisors: Lenka Zdeborová and Florent Krzakala</i><br>LAUSANNE, SWITZERLAND<br><ul style="list-style-type: none"> <li>– Developed predictive theories of feature learning in shallow and deep networks using high-dimensional probability, random matrix theory, and statistical physics.</li> <li>– Contribute more broadly to the mathematics of spin glasses and diffusion models, statistical-to-computational gaps, high-dimensional statistics, and random graphs.</li> </ul>                                           |
| 2024 – 2025 | <b>Visiting Researcher / Program Associate</b><br><i>MIT and Simons Laufer Mathematical Sciences Institute · Host: David Gamarnik</i><br>CAMBRIDGE AND BERKELEY, USA<br><ul style="list-style-type: none"> <li>– Undertook an EPFL Doc.Mobility-funded visit at MIT and participated in SLMATH's program on Probability and Statistics of Discrete Structures.</li> <li>– Developed novel techniques for analyzing first-order optimization algorithms in high dimensions and derived exact dynamical equations for Glauber dynamics in the SK model.</li> </ul> |
| SUMMER 2022 | <b>Research Intern</b><br><i>Mila · Advisors: Simon Lacoste-Julien and Yoshua Bengio</i><br>MONTREAL, CANADA<br><ul style="list-style-type: none"> <li>– Developed a theoretical framework for analyzing convergence of min-max algorithms, including the saddle-point Frank-Wolfe method.</li> </ul>                                                                                                                                                                                                                                                            |
| 2019 – 2022 | <b>Undergraduate Researcher</b><br><i>IIT Kanpur and EPFL · Collaborators: Martin Jaggi, Piyush Rai, Abhishek Kumar</i><br>INDIA AND SWITZERLAND<br><ul style="list-style-type: none"> <li>– Developed optimization methods for heterogeneous distributed learning, yielding work on implicit gradient alignment (AAAI 2022) and decentralized mixing (NeurIPS 2022 workshop).</li> <li>– Developed generalized adversarially learned inference (AAAI 2021); also co-developed a joint image-video generation model (WACV 2020).</li> </ul>                      |

## EDUCATION

|             |                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 – 2027 | <b>EPFL</b><br><i>PhD in Computer Science · expected 2027</i><br>Advisors: Lenka Zdeborová and Florent Krzakala.                                                                                                                                                    |
| 2017 – 2022 | <b>Indian Institute of Technology Kanpur</b><br><i>BTech–MTech in Computer Science and Engineering</i><br>CPI: 9.6/10; Master's CPI: 10/10. Thesis: <i>Tackling Heterogeneity in Collaborative Machine Learning</i> , co-supervised by Martin Jaggi and Piyush Rai. |

## RECOGNITION AND TALKS

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RECOGNITION        | EPFL AI Center PhD Fellowship, 2026 · Jane Street Graduate Research Fellowship (Honorable Mention), 2026 · EPFL Doc.Mobility Grant, 2024 · EDIC Fellowship, EPFL, 2022                                                                                                                                                                                                                                                                                                       |
| SELECTED TALKS     | HiLD Workshop, Banff, 2026 · Youth in High-Dimensions, Trieste, 2026 · ProbAI Scaling Laws, Warwick, 2026 · Physics of ML, Stanford, 2025 · SLMATH Seminar, Berkeley, 2025 · JMM, Seattle, 2025 · Oberwolfach, 2023                                                                                                                                                                                                                                                          |
| OTHER ACHIEVEMENTS | All India Rank 135 (of 172,024 candidates), JEE Advanced, 2017 · Aditya Birla Scholarship, 2017 · Academic Excellence Award, IIT Kanpur, 2018 and 2019                                                                                                                                                                                                                                                                                                                       |
| SERVICE            | Reviewer for Annals of Statistics, NeurIPS, ICML, ICLR, AISTATS, TMLR, and SIAM Journal on Mathematics of Data Science. Supervised several master's-level research projects at EPFL. Teaching assistant for Statistical Physics III, Statistical Physics for Optimization and Learning, Statistical Physics of Computation, and Elements of Statistics for Data Science at EPFL; Mathematics for Computer Science II/III and Introduction to Machine Learning at IIT Kanpur. |